Clock (Mod 6/2049) – General Data

1. General

The Thommen Clock is used to keep accurate time and functions as a chronometer.

2. Description

An illuminated Thommen model Q13 quartz chronometer is located on the instrument panel, to the right of the upper multi-function display. A setting knob is provided beside the clock face. The clock provides a standard 3 function, single push-button operation of a 60 minute timer (a chronometer) using start/stop/reset-to-zero in a fixed, repeating sequence. The 60 minute timer has its own minute hand (with a triangular pointer on the end) and also uses the sweep second hand for the timer function. The clock is internally illuminated and equipped with antireflective coated glass. Power for illumination is provided from the same circuit that controls all other bezel lights and is controlled by the INSTRUMENT PANEL potentiometer.

28 Volt DC power is provided from the hot battery bus through a 2 amp CLOCK AND DOME LIGHT (S1) circuit breaker on the station 332.00 circuit breaker panel (refer to 24-00-00, Electrical Power – General Data). Electrical power consumption is negligible; approximately 4 milliamps. The clock has an integral backup battery to provide power in the event of a loss of power from the hot battery bus. The backup battery will support operation of the clock for approximately one month of cumulative use.

3. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/EIL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – PANEL/PEDESTAL MOUNT UNITS CANNOT BE RELIABLY GROUNDED TO THE AIRFRAME VIA THE PANEL MOUNTING SCREWS OR DZUS FASTENERS. GROUNDING OF THESE UNITS IS ACCOMPLISHED BY USING THE SHIELD GROUNDING BRAID OR JUMPER. VERIFY PROPER ELECTRICAL BONDING AS DETAILED IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE CONNECTOR AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal of Clock

1) Ensure the aircraft electrical system is OFF.

2) Pull circuit breaker CLOCK AND DOME LIGHT (S1), on the station 332.00 circuit breaker panel and fit safety clip.

3) Gain access to the rear of the clock by removing the Upper MFD, refer to 31-11-00, Display Units – General Data.

4) Disconnect electrical connector.

Note Stow connectors of the removed components.

5) While supporting the clock from the rear of the panel, remove the (2) mounting screws.

B. Installation of Clock

1) Ensure the aircraft electrical system is OFF.

2) Ensure CLOCK AND DOME LIGHT (S1) circuit breaker is pulled and fitted with safety clip.

3) Set unit in position from behind the panel and install (2) mounting screws.

4) Reconnect electrical connector.

5) Install the Upper MFD, refer to 31-11-00, Display Units – General Data.

6) Verify that clock is operating, on battery power.

7) Remove the safety clip and close the CLOCK AND DOME LIGHT (S1) circuit breaker on the station 332.00 circuit breaker panel.

C. Removal of Battery

1) Remove the clock per Para 3.A.

2) Gently remove the battery case cover at rear of unit.

3) Remove the battery.

4) Visually inspect battery case for signs of corrosion or leakage from the battery.

D. Installation of Battery

1) Replace battery.

2) Install battery cover case, ensure case is secure.

3) Confirm clock is functioning properly.

4) Install clock as per Para 3.B.

4. Troubleshooting

A. Clock

1) Ensure CLOCK AND DOME LIGHT (S1) circuit breaker is closed.

2) Visually inspect the clock connector for damaged wiring or contacts. Ensure electrical connector on the clock is fully seated and locked.

3) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

4) If repair or service is required, send to appropriately rated avionics repair facility.

5. Servicing

A. Battery

1) Change the battery after any of the following conditions:

a) When the clock cannot maintain the correct time when aircraft power is removed.

b) There is any evidence of corrosion or leakage from the battery in the battery case.